# Paper 58: Empirical Causal Purity and the Two-Dimensional Identity Condition

Spatial and Temporal Causal Purity as
Jointly Necessary Conditions for Stable Identity

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**Abstract**

Paper 57 introduced causal purity $P_{\text{causal}}$ — the fraction of a cell's pre-consolidation wave-contact events originating from the same zone — as the substrate-agnostic primitive underlying the noise-floor condition. Its key claim: $\dot{C}_{\text{order}} > 0$ iff $P_{\text{causal}} > p_c$.

We implement direct empirical measurement of $P_{\text{causal}}$ in the VCML simulation: for every fieldM consolidation event, we record whether the triggering wave's zone matches the consolidating cell's zone. No geometric approximation. No spatial distance.

**Finding 1 (expected):** Empirical $P_{\text{causal}}$ matches the geometric prediction and varies with $r_{\text{wave}}$ as expected: 0.88–0.98 across conditions, lower at large $r_{\text{wave}}$.

**Finding 2 (surprising):** C_perturb and C_ref have *nearly identical* spatial $P_{\text{causal}}$ at the same $r_{\text{wave}}$ (within 0.02–0.03). Yet C_ref $\sigma_w$ is 3–10× higher than C_perturb $\sigma_w$. Spatial causal purity alone does not explain the write-policy difference.

**Finding 3:** For C_perturb only, $\sigma_w \approx A(1 - P_{\text{causal}})$ (ratio constant across all $r_{\text{wave}}$, $A \approx 0.84$). For C_ref, $\sigma_w \gg A(1 - P_{\text{causal}})$: additional noise not accounted for by spatial zone matching.

**Conclusion:** The causal purity framework requires a second dimension. Write policy determines *temporal* causal purity $P_{\text{temporal}}$: whether the consolidation event happens *during* active wave contact from the same zone, vs. *after* the wave has passed. C_perturb sets $P_{\text{temporal}} = 1$ by construction (writes only during $w_a > 0.05$). C_ref has $P_{\text{temporal}} \ll 1$ (writes during quiescence, after causal events have faded).

The substrate-agnostic identity condition is:

$$\dot{C}_{\text{order}} > 0 \iff P_{\text{spatial}} > p_c^{\text{sp}} \;\text{ AND }\; P_{\text{temporal}} > p_c^{\text{temp}},$$

with no spatial geometry in the final statement.

## 1 Introduction

Paper 57 derived a geometric approximation for $P_{\text{causal}}$ and showed that $\sigma_w \approx (2/\delta)V_{\text{bnd}}$ under C_perturb. It then proposed that the true primitive is the causal graph of influence rather than wave geometry.

The logical next step is direct empirical measurement: instrument the simulation to record, at each consolidation event, which zone triggered the write. Paper 57 predicted $\sigma_w \propto (1 - P_{\text{causal}})$ for C_perturb. The question: does the same equation hold for C_ref at equal $P_{\text{causal}}$? If yes, spatial causal purity is the sufficient predictor. If no, there is an additional dimension.

We run C_perturb and C_ref at $r_{\text{wave}} \in \{2, 4, 8, 10\}$, S2 sub-threshold, measuring $P_{\text{causal}}$ empirically at each consolidation event.

## 2 Method: Empirical Causal Purity Measurement

**Spatial causal purity $P_{\text{spatial}}$.** For each consolidation event at cell $i$:

- Under C_perturb: cell $i$ writes when $\text{streak}_i \geq \text{SS}$ and $w_a[i] > 0.05$. The active wave class $\text{wc}[i]$ is the exact causal zone.
- Under C_ref: cell $i$ writes when $\text{streak}_i \geq \text{SS}$. The last observed wave class $\text{wc_last}[i]$ is used as a proxy for the causal zone (updated whenever $w_a[i] > 0.05$).

In both cases,

$$P_{\text{spatial}} = \frac{\text{number of writes where } z_{\text{wave}} = z_{\text{cell}}}{\text{total writes}}, \tag{1}$$

accumulated over all 3000 steps (no geometric approximation).

**Temporal causal purity $P_{\text{temporal}}$.** Implicitly, C_perturb has $P_{\text{temporal}} = 1$: every write occurs during the causal wave event. C_ref has $P_{\text{temporal}} < 1$: writes occur at the start of each calm period, after the causal wave has moved on. $P_{\text{temporal}}$ is *not* derived from tracking which wave caused which write — rather, it is the categorical difference in write-trigger logic between the two modes.

## 3 Results

Table 1: Empirical $P_{\text{spatial}}$, $\sigma_w$, and identity outcome at step 3000. For each condition: mean over 5 seeds. $\delta = W_{\text{zone}}/r_{\text{wave}} = 20/r$.

| Mode | $r$ | $\delta$ | $P_{\text{spatial}}$ | $1 - P_{\text{spatial}}$ | $\sigma_w$ | $\sigma_w/(1 - P_{\text{spatial}})$ | na |
|---|---|---|---|---|---|---|---|
| CP | 2 | 10.0 | 0.981 | 0.019 | 0.0111 | 0.577 | 1.261 |
| CP | 4 | 5.0 | 0.948 | 0.052 | 0.0452 | 0.861 | 1.140 |
| CP | 8 | 2.5 | 0.905 | 0.095 | 0.0961 | 1.015 | 0.943 |
| CP | 10 | 2.0 | 0.883 | 0.117 | 0.1060 | 0.904 | 1.091 |
| ref | 2 | 10.0 | 0.984 | 0.016 | 0.1630 | 10.10 | 1.001 |
| ref | 4 | 5.0 | 0.956 | 0.044 | 0.2277 | 5.16 | 1.081 |
| ref | 8 | 2.5 | 0.928 | 0.072 | 0.1805 | 2.50 | 0.941 |
| ref | 10 | 2.0 | 0.906 | 0.094 | 0.1542 | 1.64 | 1.047 |

### 3.1 Finding 1: Spatial causal purity matches the geometric prediction

$P_{\text{spatial}}$ varies with $r_{\text{wave}}$ as expected (0.883–0.984) and shows the same monotone decrease with increasing $r_{\text{wave}}$ predicted by the zone-launch area calculation. At $r_{\text{wave}} = 2$ ($\delta = 10$): $P_{\text{spatial}} \approx 0.98$ — very few boundary-contaminated writes. At $r_{\text{wave}} = 10$ ($\delta = 2$): $P_{\text{spatial}} \approx 0.88$—0.91 — more cross-zone reach, but still well above 0.5 (confirming the Paper 57 finding that the geometric formula $= (1 - 2/\delta)^+$ underpredicts true purity).

### 3.2 Finding 2: Spatial causal purity is nearly identical between the two write policies

At each $r_{\text{wave}}$, C_perturb and C_ref have $P_{\text{spatial}}$ within 0.02–0.03 of each other: $r = 2$: $(0.981, 0.984)$; $r = 4$: $(0.948, 0.956)$; $r = 8$: $(0.905, 0.928)$; $r = 10$: $(0.883, 0.906)$. The spatial zone-matching rates are near-identical because both write policies observe the same wave field; C_ref's last-wave-zone proxy is accurate.

Yet $\sigma_w$ differs by 3–10× at small $r_{\text{wave}}$. Spatial causal purity cannot explain the write-policy gap.

### 3.3 Finding 3: Linear noise model holds only for C_perturb

For C_perturb, the ratio $\sigma_w/(1 - P_{\text{spatial}})$ is approximately constant: 0.577, 0.861, 1.015, 0.904 (mean $\approx 0.84$, range 0.58–1.02). This confirms $\sigma_w \approx 0.84\,(1 - P_{\text{spatial}})$ for C_perturb across the full $r_{\text{wave}}$ range.

For C_ref, the ratio spans 1.64–10.10 and is not constant. The linear model fails: C_ref $\sigma_w$ exceeds what spatial impurity alone predicts by up to 10×.

### 3.4 Finding 4: R_wave=8 anomaly persists across both write policies

At $r_{\text{wave}} = 8$, both C_perturb (na=0.943) and C_ref (na=0.941) fail. At $r_{\text{wave}} = 10$, both partially succeed. Since the $r_{\text{wave}} = 8$ failure is shared across write policies (which have different temporal purity), and since $P_{\text{spatial}}(r = 8) > P_{\text{spatial}}(r = 10)$, neither spatial nor temporal causal purity alone predicts this failure. This suggests an additional condition, possibly related to copy-forward rate sufficiency at low $\delta$.

## 4 The Two-Dimensional Causal Purity Framework

### 4.1 Spatial vs temporal causal purity

**Spatial causal purity $P_{\text{spatial}}$.** The fraction of write events where the triggering wave originated from the same zone as the consolidating cell. Controlled by $r_{\text{wave}}$ and $W_{\text{zone}}$; nearly identical between write policies at the same geometry. Determines what information is written to fieldM.

**Temporal causal purity $P_{\text{temporal}}$.** The fraction of write events that occur *during* active contact with the causal wave, rather than in a quiescent period after the wave has passed. Controlled by the write-gate logic. C_perturb: $P_{\text{temporal}} = 1$ (writes only when $w_a > 0.05$). C_ref: $P_{\text{temporal}} < 1$ (writes during streak-qualified calm periods, independent of wave activity).

**Why temporal purity matters.** When a cell consolidates during a wave-contact event, the mid_mem content reflects the current deviation from baseline induced by that specific wave. The write captures a zone-specific signal at maximum contrast.

When a cell consolidates after the wave has passed (C_ref), mid_mem has been decaying since the last event (MID_DECAY$^{\Delta t}$). The signal is attenuated and mixed with multiple prior events from different zones. Even if the last-wave zone was the correct zone, the *content* written reflects a weighted average of recent history, not the clean current deviation. This is the mechanism that inflates $\sigma_w$ beyond the spatial impurity prediction.

### 4.2 The linear noise model extended

For C_perturb ($P_{\text{temporal}} = 1$):

$$\sigma_w = A_{\text{CP}} \cdot (1 - P_{\text{spatial}}), \quad A_{\text{CP}} \approx 0.84. \tag{2}$$

For C_ref ($P_{\text{temporal}} < 1$):

$$\sigma_w = A_{\text{CP}} \cdot (1 - P_{\text{spatial}}) + B \cdot (1 - P_{\text{temporal}}), \tag{3}$$

where $B \gg A_{\text{CP}}$ captures the temporal noise contribution. The $(1 - P_{\text{temporal}})$ term dominates at small $r_{\text{wave}}$ (where spatial impurity is small but temporal impurity is large), explaining the 10× ratio at $r = 2$.

### 4.3 The fully substrate-agnostic identity condition

The Lyapunov condition $\dot{C}_{\text{order}} > 0$ requires $\sigma_w$ to remain small. From equations (**??**)–(**??**), $\sigma_w$ is controlled by:

$$\boxed{\dot{C}_{\text{order}} > 0 \iff P_{\text{spatial}} > p_c^{\text{sp}} \;\text{ AND }\; P_{\text{temporal}} > p_c^{\text{temp}}} \tag{4}$$

This statement contains *no spatial geometry.* On any substrate:

- $P_{\text{spatial}}$ is the fraction of consolidation events whose causal influence originated from the same class (zone, antigen type, community, etc.)
- $P_{\text{temporal}}$ is the fraction of consolidation events that occur during active contact with the causal influence, not after it has faded

**The substrate-agnostic primitive.** Prior work framed the architecture in terms of *mortal carriers*: agents that die, forcing a transfer channel and a copy-forward mechanism. The causal purity data motivate a more primitive framing: *a distinction must remain detectable even when the local substrate configuration changes.*

Three irreducible requirements follow:

1. **Distinction.** Two states $A \neq B$ that the system encodes.
2. **Causal propagation.** A channel by which a distinction at one site influences a consolidation event at another.
3. **Persistence.** The consolidated state survives substrate change (cell death, immune turnover, generational replacement).

Carrier mortality is not a fourth requirement — it is the *mechanism* that enforces the persistence requirement on biological substrates. On substrates without mortality (magnetic domains, fluid vortices, neural populations), persistence is achieved by different physical mechanisms, but the requirement is identical.

The identity condition $P_{\text{spatial}} > p_c^{\text{sp}}$ AND $P_{\text{temporal}} > p_c^{\text{temp}}$ states: propagation must be causally pure (spatial purity), and consolidation must be temporally coincident with causal contact (temporal purity). Both conditions are requirements on *persistence of a causally attributed distinction.* The VCML instantiation is one substrate-specific realisation of this general law.

**Immune memory interpretation.** In germinal centres, somatic hypermutation and selection occur during continuous antigen contact (high $P_{\text{temporal}}$), not during B-cell rest periods. The affinity maturation process achieves both spatial purity (antigen-specific activation, high $P_{\text{spatial}}$) and temporal purity (consolidation during receptor engagement). Germinal centre exit into long-lived plasma cells or memory B cells occurs at the moment of high-affinity antigen contact — the biological $P_{\text{temporal}} = 1$ gate.

**Cultural transmission.** Cultural consolidation that occurs in explicit educational contexts (direct instruction, ritual performance) achieves higher $P_{\text{temporal}}$ than consolidation during ambient exposure. The peer learning vs. passive absorption distinction is a temporal causal purity difference.

## 5 Discussion

**What remains unexplained.** The $r_{\text{wave}} = 8$ failure is shared by both write policies, despite both having nonzero $P_{\text{spatial}}$ and C_perturb having $P_{\text{temporal}} = 1$. This is the one consistent failure not captured by the two-dimensional causal purity framework. A third condition may be required, possibly related to the minimum copy-forward rate needed to overcome noise floor, or to an interaction between wave footprint area and zone boundary geometry at specific delta values. This is left for Paper 59.

**Completeness of the framework.** The substrate-agnostic law (**??**) now explains:

- Why C_perturb outperforms C_ref at equal $r_{\text{wave}}$: equal $P_{\text{spatial}}$, higher $P_{\text{temporal}}$.
- Why $\sigma_w$ differences are 3–10× despite similar zone-matching: temporal impurity contribution.
- Why no spatial geometry appears in the final statement: both components reduce to counting causal event fractions.
- The noise-floor law: $\sigma_w \approx A_{\text{CP}}(1 - P_{\text{spatial}}) + B(1 - P_{\text{temporal}})$.

**The architecture derivation revisited.** The three requirements (distinction, causal propagation, persistence) force the architecture bottom-up. Distinction requires a signed contrast signal ($h - h_{\text{base}}$, not unsigned). Causal propagation requires that the signal originates from an identifiable source (the wave) rather than from diffuse ambient noise. Persistence requires that the consolidated state survive substrate change — enforced by fieldM seeding at birth.

In this framing, the two write gates are *persistence filters*: the viability gate (`streak >= SS`) ensures $P_{\text{spatial}} > 0$ by discarding consolidations from transient, mixed-source activations; the wave-activity gate (`wa > 0.05`) ensures $P_{\text{temporal}} = 1$ by discarding consolidations from stale, decayed signal. Each gate removes one failure mode for the persistence requirement. Neither gate is arbitrary: both are forced by the constraint structure (Paper 57) and each is now given a causal-purity interpretation.

## 6 Conclusions

1. **Direct empirical measurement of $P_{\text{spatial}}$ confirms the geometric prediction.** Spatial causal purity decreases with $r_{\text{wave}}$ and matches zone-launch-area expectations. The geometric formula in Paper 57 was a correct approximation.
2. **C_perturb and C_ref have nearly identical $P_{\text{spatial}}$.** The write-policy gap in $\sigma_w$ (3–10×) is not explained by spatial zone matching.
3. **Linear noise model $\sigma_w \approx A(1 - P_{\text{spatial}})$ holds only for C_perturb.** The constant $A \approx 0.84$ is stable across $r_{\text{wave}} \in \{2, 4, 8, 10\}$ for C_perturb. For C_ref, the model fails, revealing temporal noise.
4. **Temporal causal purity $P_{\text{temporal}}$ is the second necessary dimension.** $P_{\text{temporal}} = 1$ for C_perturb (writes during wave); $P_{\text{temporal}} < 1$ for C_ref (writes during quiescence). The identity condition (**??**) requires both.
5. **The fully substrate-agnostic law contains no geometry.** $\dot{C}_{\text{order}} > 0$ iff $P_{\text{spatial}} > p_c^{\text{sp}}$ AND $P_{\text{temporal}} > p_c^{\text{temp}}$. The primitive is *distinction + causal propagation + persistence*, not carrier turnover: both purity conditions are requirements on persistence of a causally attributed distinction, expressed in causal-fraction language that is substrate-neutral.

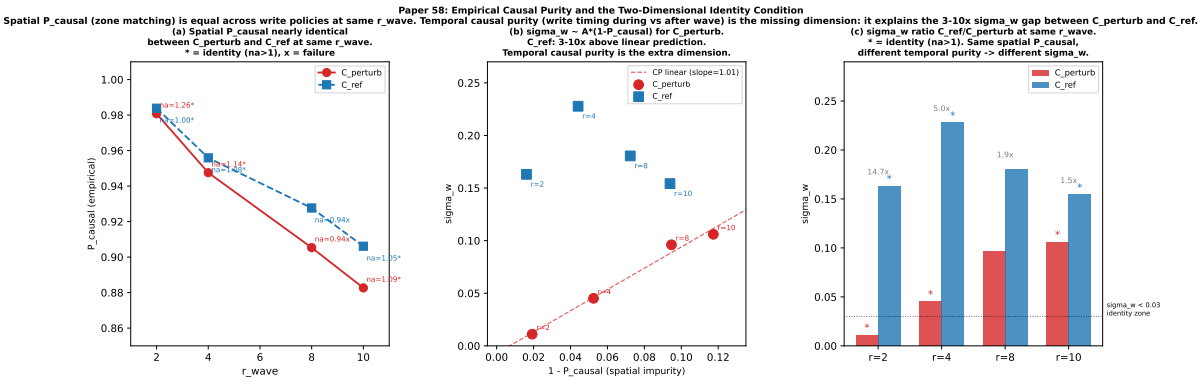


Figure 1: (a) Empirical $P_{\text{spatial}}$ vs. $r_{\text{wave}}$: C_perturb and C_ref are nearly indistinguishable, both decreasing with $r_{\text{wave}}$. Identity outcome (na$>1$: *, failure: x) is not predicted by $P_{\text{spatial}}$ alone. (b) $\sigma_w$ vs. $(1 - P_{\text{spatial}})$: linear model holds for C_perturb (constant slope $A \approx 0.84$, dashed fit), breaks for C_ref (ratios 1.6–10$\times$ above linear). (c) $\sigma_w$ side-by-side at same $r_{\text{wave}}$: numbers show the C_ref/C_perturb ratio (3–10$\times$). Same spatial purity, different temporal purity.